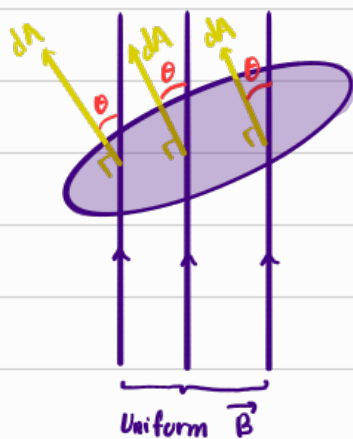


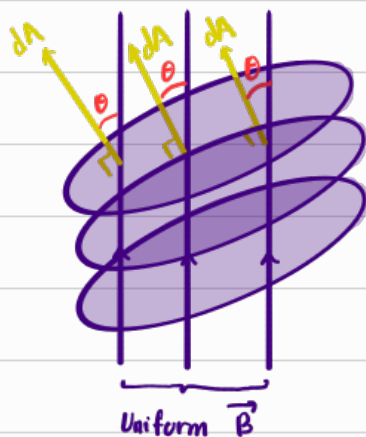
Magnetic Flux, ϕ_B



$$\begin{aligned}\phi_B &= \int \vec{B} \cdot d\vec{A} \\ &= \vec{B} \int d\vec{A} \\ &= \vec{B} \cdot \vec{A}\end{aligned}$$

$$\phi_B = BA \cos \theta$$

Magnetic Flux Linkage, Φ_B



$$\Phi_B = \sum \phi_B = N BA \cos \theta$$

No. of loops

Faraday's Law [Simplified]

$$\mathcal{E}_{\text{induced}} \propto \frac{d\phi_B}{dt}$$

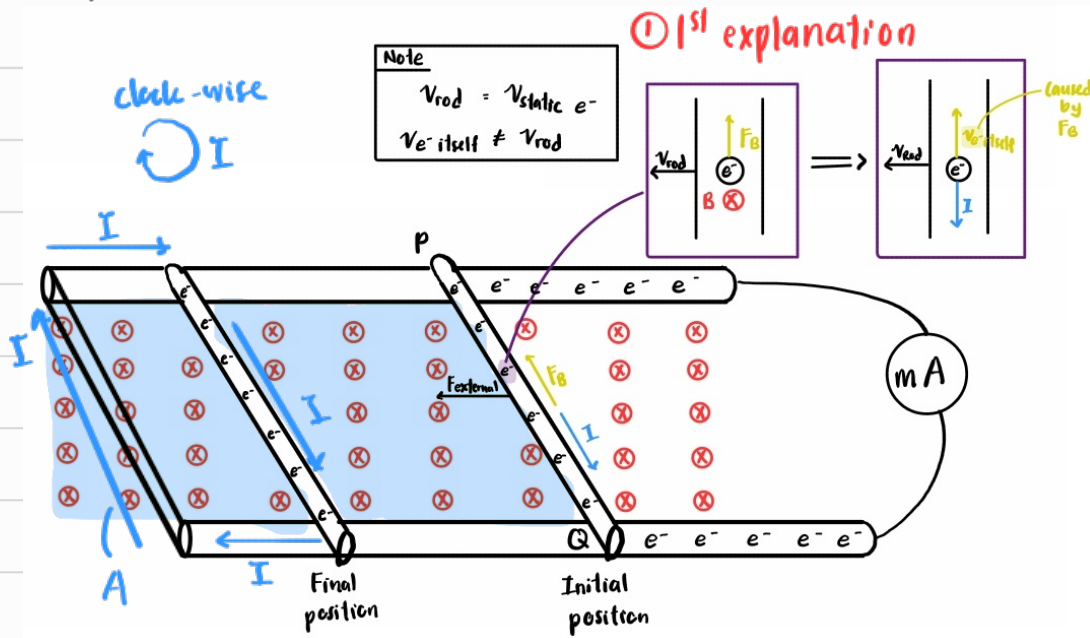
$$(I_{\text{induced}})R \propto \frac{d\phi_B}{dt}$$

Lenz Law

$$\mathcal{E}_{\text{induced}} = - \frac{d\phi_B}{dt}$$

The direc. of induced emf produces $\vec{B}_{\text{induced}}$ that opposes the change causing it.

Explanation of Lenz Law / Direc. of Induced



② 2nd explanation

As PQ moves, $A \downarrow$

$\rightarrow \phi_B \downarrow$

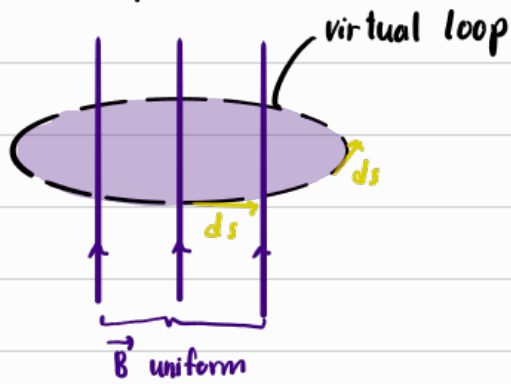
$\rightarrow$ Since there is a $\Delta \phi_B$, $\mathcal{E}_{induced}$ will be produced [Faraday's Law]

$\rightarrow \mathcal{E}_{induced}$ will be produced in a way that produces $\vec{I}_{induced}$ in the direc. that produces a $\vec{B}_{induced}$ to oppose the $\Delta \phi_B$ [Lenz Law]

$\rightarrow$ Since $\phi_B \downarrow$, $\vec{B}_{induced}$ have to point in the same direc. as $\vec{B}_{external}$ to increase back ϕ_B

$\rightarrow$ Thus, using Right Thumb Rule, we know that Induced flow is the clock-wise direction.

Relationship btwn $\vec{E}$ & $\vec{B}$



$$\mathcal{E}_{\text{induced}} = - \frac{d\Phi_B}{dt}$$

* Note :

Faraday's Law actually has 2 forms

$$\begin{aligned} \textcircled{1} \mathcal{E}_{\text{induced}} &= - \frac{d\Phi_B}{dt} & \mathcal{E} = \oint \vec{E} \cdot d\vec{s} \text{ can} \\ \textcircled{2} \oint \vec{E}_{\text{induced}} \cdot d\vec{s} &= - \frac{d\Phi_B}{dt} & \text{be derived from} \\ & & \text{Max-Well eqns.} \end{aligned}$$

So technically $\Delta\Phi_B$ doesn't directly produce $\mathcal{E}$.

The flow is as follows :

$$\Delta\Phi_B \longrightarrow \mathcal{E}_{\text{induced}} \longrightarrow \oint \vec{E}_{\text{induced}} \cdot d\vec{s} \longrightarrow I_{\text{induced}} \longrightarrow B_{\text{induced}}$$

$$\mathcal{E}_{\text{induced}} = \oint \vec{E}_{\text{induced}} \cdot d\vec{s}$$

In other words,

The total push of e^- , $\mathcal{E}$ around the loop

= The sum of all tiny pushes around the loop.

* Even on a virtual loop, there will be

$\mathcal{E}_{\text{induced}}$ & $\oint \vec{E}_{\text{induced}} \cdot d\vec{s}$, but no I_{induced} cuz there

are no free moving charge carriers. Thus there is also no B_{induced} .

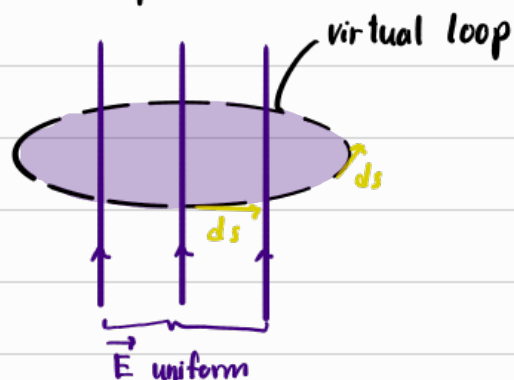
* Do not mix up

① Electrostatic Electric Fields, $\mathcal{E}$ from charges

② Induced Electric Fields, $\mathcal{E}_{\text{induced}}$ from $\Delta\Phi_B$ [Form Loops]

$$\begin{aligned} \therefore \oint \vec{E} \cdot d\vec{s} &= - \frac{d\Phi_B}{dt} \\ \mathcal{E} \oint ds &= - \frac{d(B(t) A \cos\theta)}{dt} \\ \mathcal{E} \oint ds &= - (A \cos\theta) \frac{d}{dt} B(t) \end{aligned}$$

Relationship btwn $\vec{B}$ & $\vec{E}$



When ② has no I_{enclosed}

$$\rightarrow \oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

This means a $\Delta\Phi_E$ induces a $\vec{B}$

* Note:

Ampere's Law actually has 2 forms

$$\textcircled{1} \oint \vec{B} \cdot d\vec{s} = \mu_0 I_{\text{enclosed}} \quad [\text{Simplified}]$$

$$\textcircled{2} \oint \vec{B} \cdot d\vec{s} = \mu_0 [I_{\text{enclosed}} + I_{\text{displacement}}] \\ = \mu_0 [I_{\text{enclosed}} + \epsilon_0 \frac{d\Phi_E}{dt}] \quad [\text{Maxwell}]$$

A $\Delta\Phi_E$ creates ^{Imaginary} $I_{\text{displacement}}$ that behaves as a real current in generating $\vec{B}$ but does not involve flow of actual charges

The flow is as follows,

$$\Delta\Phi_E \rightarrow I_{\text{imaginary}} \rightarrow \vec{B}$$

* Do not mix up

① Static Magnetic Fields, $\vec{B}$ from steady current

② Induced Magnetic Fields, $\vec{B}_{\text{induced}}$ from $\Delta\Phi_E$

Since steady current & $\Delta\Phi_E$ produces $\vec{B}$, Maxwell improved on Ampere's Law by adding the $I_{\text{displacement}}$ term.

$$\therefore \oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} \\ B \oint ds = \mu_0 \epsilon_0 \left[\frac{d[E(t)A]}{dt} \right] \\ B \oint ds = A \mu_0 \epsilon_0 \left[\frac{d}{dt} E(t) \right]$$

